

Transportation Telemetry Predictive Timing Platform

Synthetic train telemetry, lakehouse pipeline, and neural network delay prediction

Project question

Can a real-time transportation telemetry platform predict train delay timing and support predictive maintenance decision-making using sensor, location, and operational event data?

Overview **Metrics** Figures Architecture

Accuracy	F1	AUC	MAE minutes	RMSE minutes
70.72%	0.00%	55.47%	14.99	16.71

	actual_delay_risk	predicted_delay_risk	predicted_delay_probability	actual_delay_minutes	predicted_delay_minutes
0	1	0	0.3902	21.8017	0.8655
1	1	0	0.3829	29.257	1.2574
2	1	0	0.4039	24.1407	0.614
3	0	0	0.3805	18.3709	0.8555
4	0	0	0.3875	16.1714	1.0599
5	0	0	0.4116	14.2534	1.0041
6	0	0	0.338	15.7116	1.6196
7	0	0	0.4105	17.3818	0.6934
8	0	0	0.3485	6.6393	1.1895
9	1	0	0.4255	24.5865	1.1247